

Challenge (Habanero)

- Q1.** A recipe for baking bread requires $2 \cdot 3.75$ cups of flour to make 15 loaves. What is the unit rate of flour per loaf of bread?
(A) 0.5 cups
(B) 0.75 cups
(C) 1 cup
(D) 1.25 cups
(E) 1.5 cups
- Q2.** If a water tank is being filled at a rate of $4.2 \cdot 10^2$ liters per minute, how many liters will it hold after 15 minutes?
(A) $6.3 \cdot 10^3$ liters
(B) $6.5 \cdot 10^3$ liters
(C) $7.2 \cdot 10^3$ liters
(D) $8.1 \cdot 10^3$ liters
(E) $9.0 \cdot 10^3$ liters
- Q3.** The cost of a construction project varies directly with the number of workers. If a project that requires 18 workers costs 36000, what is the cost per worker?
(A) 1500
(B) 2000
(C) 2500
(D) 3000
(E) 3500
- Q4.** A cake recipe calls for $2\frac{1}{4}$ cups of sugar for every 2 cakes. What is the unit rate of sugar per cake?
(A) $1\frac{1}{8}$ cups
(B) $1\frac{1}{4}$ cups
(C) 1.125 cups
(D) 1.25 cups
(E) 1.5 cups
- Q5.** The volume of a rectangular prism varies directly with its length. If a prism with a length of 8 feet has a volume of 48 cubic feet, what is the volume of a prism with a length of 12 feet?
(A) 48 cubic feet
(B) 72 cubic feet
(C) 90 cubic feet
(D) 108 cubic feet
(E) 120 cubic feet
- Q6.** The speed of a car is $45 \cdot 10^3$ meters per 4 hours. What is the unit rate of distance per hour?
(A) $5.625 \cdot 10^3$ meters
(B) $6.25 \cdot 10^3$ meters
(C) $7.5 \cdot 10^3$ meters
(D) $8.75 \cdot 10^3$ meters
(E) $11.25 \cdot 10^3$ meters
- Q7.** A bakery sells 480 loaves of bread per day. If each loaf requires 0.75 cups of flour, how many cups of flour are used in a day?
(A) 240 cups
(B) 288 cups
(C) 320 cups
(D) 360 cups
(E) 4320 cups
- Q8.** A pipe can fill a tank at a rate of 12 cubic feet per minute. If the tank has a capacity of $2.4 \cdot 10^3$ cubic feet, how many minutes will it take to fill the tank?
(A) 150 minutes
(B) 180 minutes
(C) 200 minutes
(D) 220 minutes
(E) 240 minutes

Open Ended Questions (FRQ)

- Q9.** A construction project requires 15 workers to complete 3 units of work in 8 hours. If the number of workers is increased to 20, how many hours will it take to complete 5 units of work? Show all work and explain your reasoning.
- Q10.** A chef is making a sauce that requires a ratio of 2 : 3 of two ingredients. If the chef has 12 cups of one ingredient, how many cups of the other ingredient are needed? If the amount of one ingredient is increased to 18 cups, how will the amount of the other ingredient change? Use mathematical models to support your answer.

Chef's Tips

- Read each question carefully before answering.
- Use a pencil to fill in the answer circles.
- Check your work before submitting your test.